

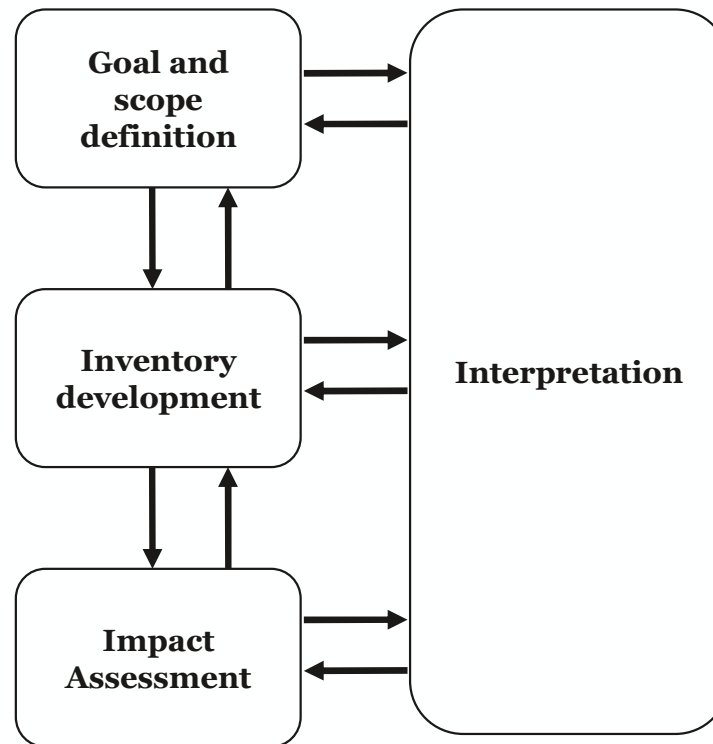


ENEL301 Fundamentals of Engineering
Economics and Management

Life cycle assessment #2

Dr Steve Pudney
Erkine Fellow

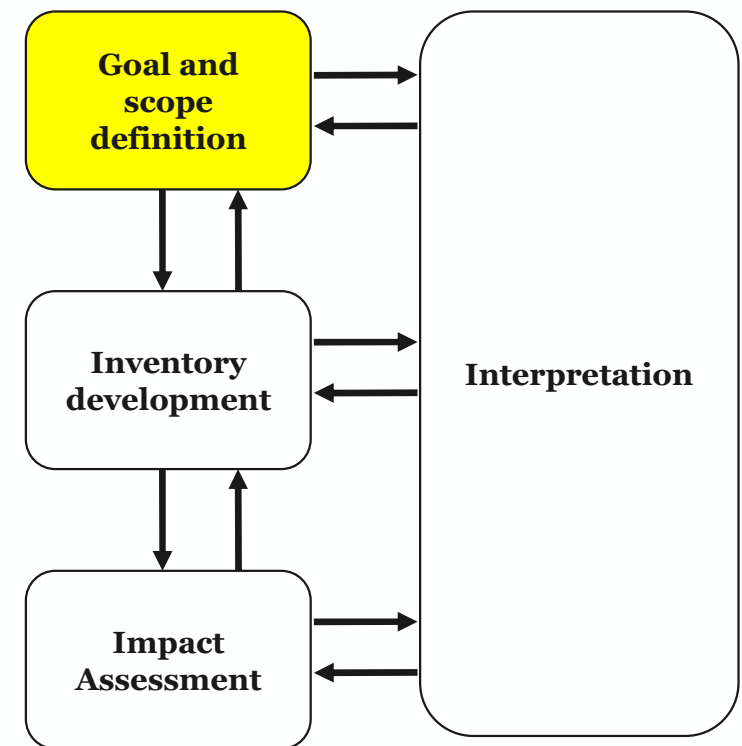
Recap....



Source: ISO 14040:2006

Goal and scope definition

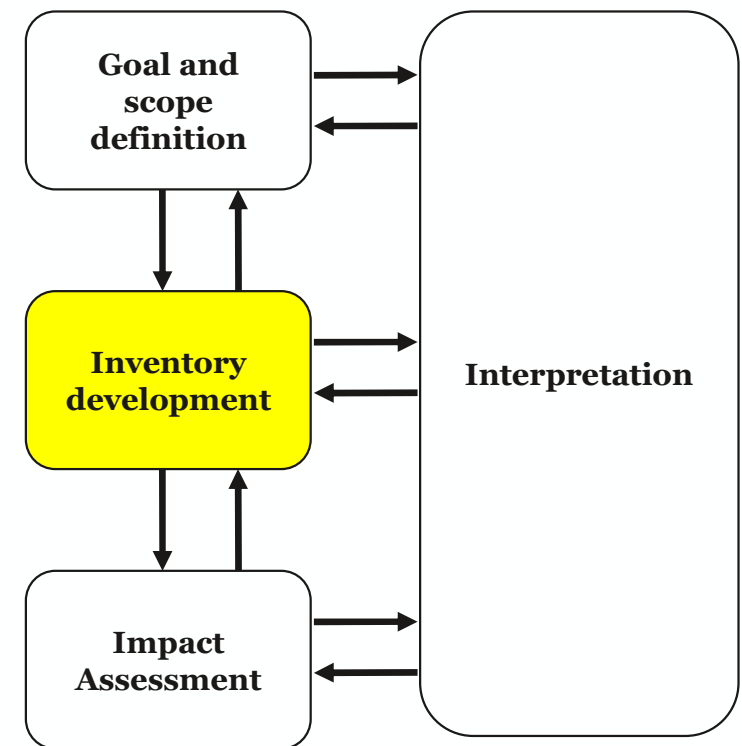
- Frame the study
- Functional unit
- System boundary
- Identify environmental indicators
- Identify data requirements



Source: ISO 14040:2006

Inventory development

- LCA project breadth of data
- Foreground data
 - Data that the original manufacturer supplies, e.g. mass of packaging, energy inputs for a process, emissions. **Data comes from the business.**
- Background data
 - Data that the LCA practitioner supplies
 - e.g. electricity generation, transport, production of common materials
 - **Typically, from LCA databases**



Source: ISO 14040:2006

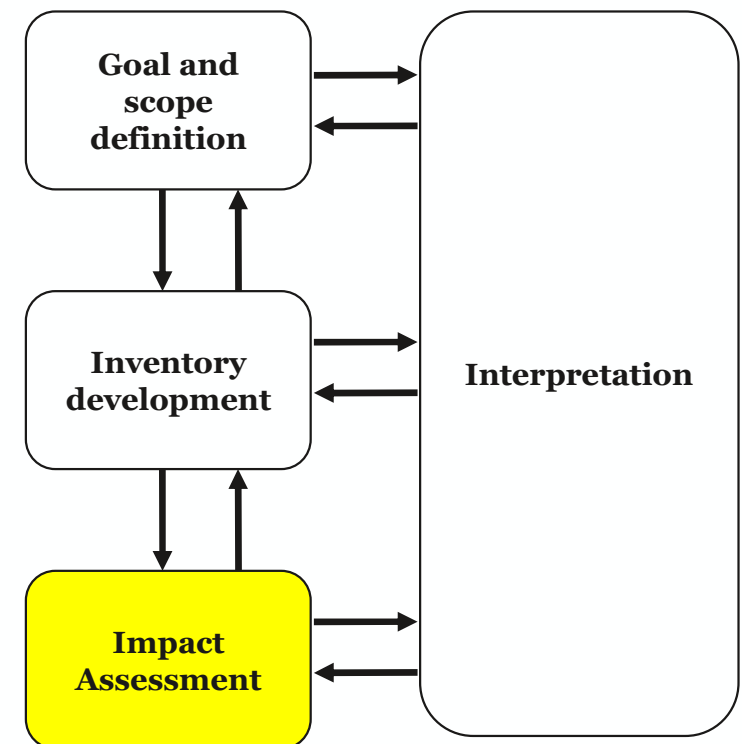
Impact assessment

Must:

- Characterise: calculate impact assessment

Optional:

- Normalise: contribution relative to a known baseline (e.g. national load)
- Weighting of indicators to fewer scores
 - Subjective → need to calculate for different weightings



Source: ISO 14040:2006

Interpretation

Iterate back to other steps to:

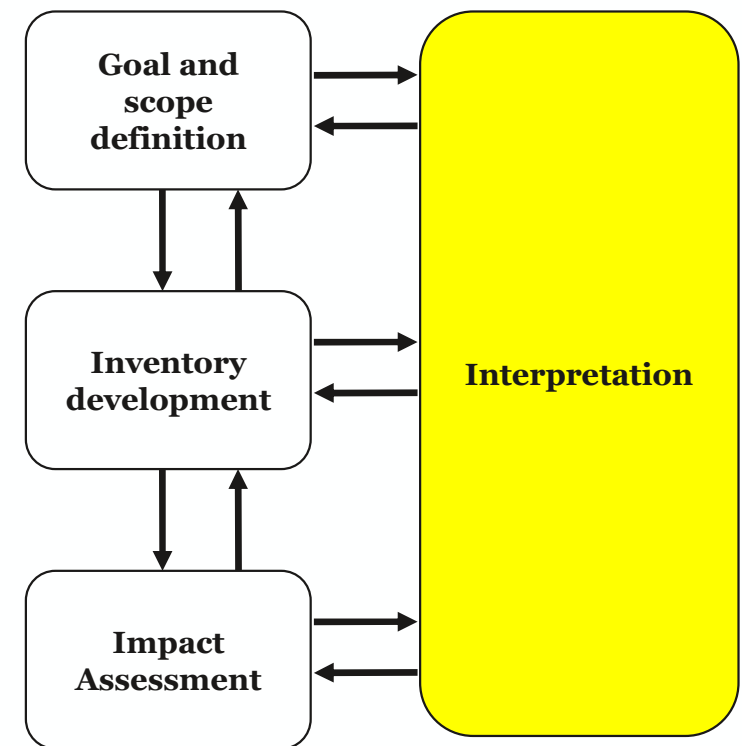
- Identify the reasons for your results → '5 whys?'
- Check methods inc. assumptions, data quality

Sensitivity studies

- The effect of key assumptions on your outcomes

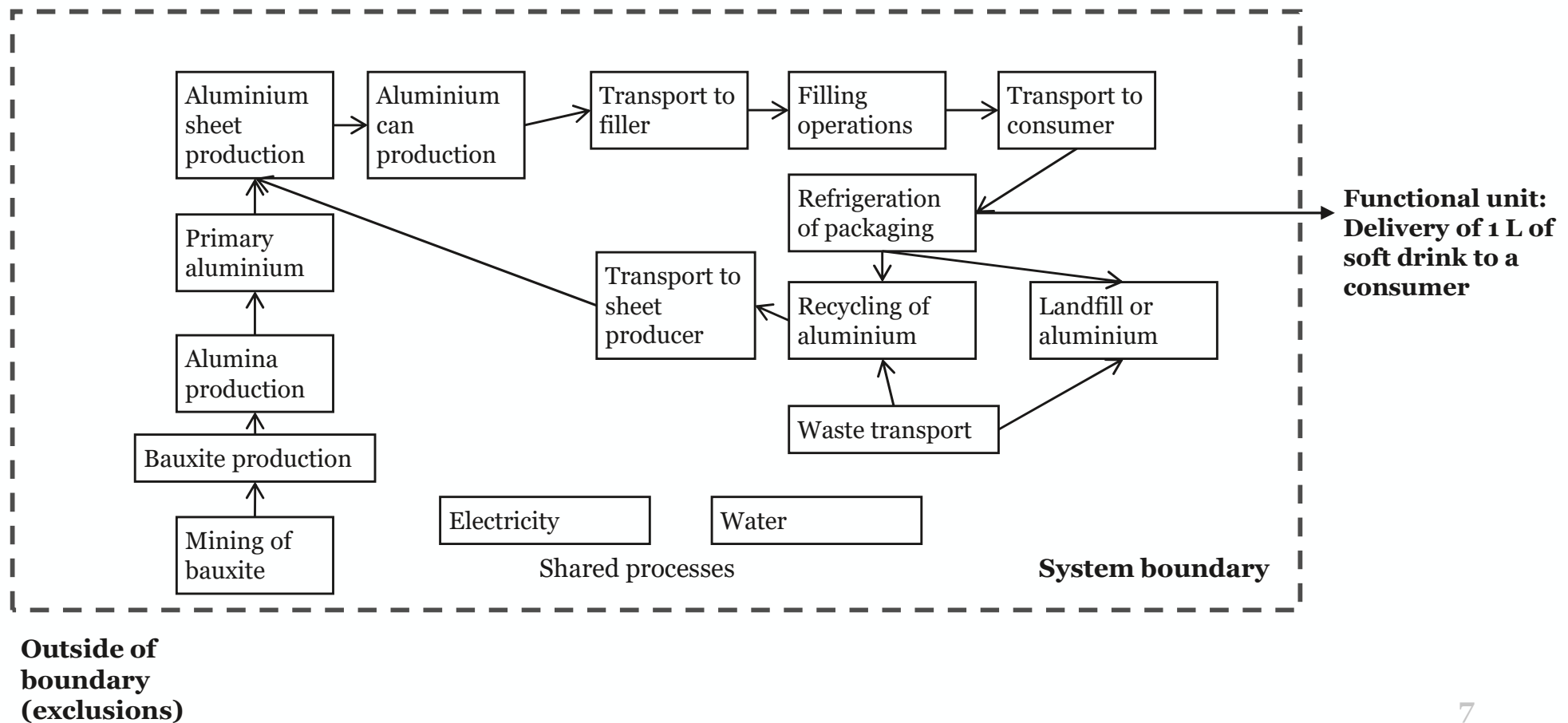
Uncertainty studies

- The effect of data uncertainty on outcomes → confidence in your claims
- Monte Carlo simulations



Source: ISO 14040:2006

Recall the production process for aluminium drink container sufficient for 1 litre of drink



Install OpenLCA



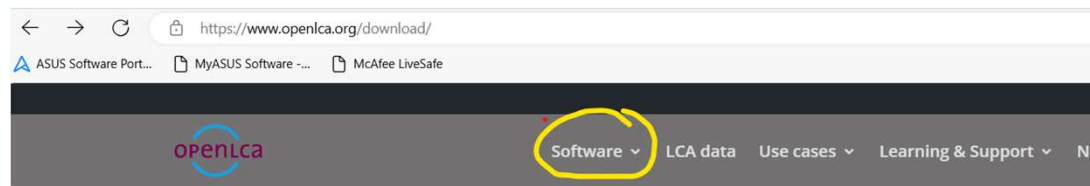
🔍 OpenLCA



Google Search

I'm Feeling Lucky

Download OpenLCA



Visit our public support platform [ask.openLCA](#), – a question-and-answer (Q&A) website on Life Cycle Assessment and also the official support platform for [openLCA](#), [openLCA nexus](#) and the [LCA Collaboration Server](#).

[ask.openLCA](#)

Downloads



openLCA

Stable versions for Windows, macOS, and Linux, development versions (if any), some previous versions, sources, and a changelog for the openLCA LCA and sustainability modeling software are provided below.

Since openLCA 2.0.3 you do not need to download the supplementary UMFPACK library for accelerated calculations. We now include the Intel Math Kernel Library (MKL) which is a highly performant, optimized, and efficient library for mathematical and scientific computations. openLCA 2.0.3 with the previous math libraries can still be downloaded under the *Previous* tab, within the *blas* folder.

[Release notes for openLCA 2.0](#) can be found in our blog.

The most recent version (non-beta) is version 2.3.0 from September 30, 2024. For further updates for later versions, please refer to the changelog below.



openLCA Nexus

Your source for LCA and sustainability data.

Databases

EN15804 add-on
ecoinvent
Circularity Package
The Evah Pigments Database
LCA Commons (complete)
IDEMAT
Carbon Minds
PLEX
UVEK LCI Data
OzLCI2019
Environmental Footprints
idea
Agri-footprint
exiobase
ARVI
Agribalyse
Circularity Food Package
soca
EuGeos' 15804-IA

All

Free databases

For purchase databases



OzLCI2019

The Evah OzLCI2019 Free Database has been created by our Australian partner The Evah Institute. It covers Australasian regional supply including imports and has been developed by using openLCA. It can be used in combination with the latest version of the openLCA impact assessment methods (LCIA methods), which is available under downloads and free of charge.

Browse



Environmental Footprints

The Environmental Footprint (EF) database is now provided under the guidance of the European Commission and its Joint Research Centre in its

update

The screenshot displays the openLCA 2.3.0 software interface. The title bar reads "openLCA 2.3.0 - Aluminium drink container". The menu bar includes "File", "Database", "Tools", and "Help". The left sidebar, titled "Navigation", shows a tree structure under "ENEL301":

- Aluminium drink container (highlighted with a yellow box and a red arrow pointing to it)
- Projects
- Product systems
- Processes
- Flows
- EPDs
- Results
- Indicators and parameters
- Background data (highlighted with a red arrow pointing to it)
 - Flow properties
 - Unit groups
 - Currencies
 - Actors
 - Sources
 - Locations
 - Mapping files

The main window shows a "Welcome" screen with the openLCA logo and a cityscape background. The text on the screen includes:

- Getting started
- What's new in openLCA 2
- Extensive documentation
- Community Q&A
- Databases, case studies, impact methods
- Certified trainings, support
- Tools to get more out of openLCA
- Working with the developers of openLCA

Additional text on the right side of the screen:

Trainings on LCA, social LCA, sustainability and of course openLCA are available on a regular basis, provided by GreenDelta and also by other certified trainers worldwide. They are posted and can be booked on Nexus.

Software and modelling support is available as well, it can be booked via Nexus, contact us for larger packs or inquiries.

The "GreenDelta" logo is visible in the bottom right corner of the main window.



openLCA Nexus

Your source for LCA and sustainability data.

Overall providing ~ 300,000 data sets.

PSILCA v3.1 now available!

e.g. for Switzerland: 67510 data sets found.

openLCA manual update

by Sarah Serafini | Oct 8, 2024 | Life Cycle Impact Assessment

With the release of openLCA 2.3, we also provide an updated version of openLCA handbook manual with the following changes: Adapted to changes coming along with the release of openLCA 2.3 Novel chapter on "Dependent Parameters" Expanded...

Software update release: openLCA 2.3.0

by Julia Cilleruelo | Oct 7, 2024 | openLCA releases

Last week we released version 2.3.0 of openLCA. We would like to encourage users to download this newest update. The changes from the previous version are: Support for the newest CS version & improvements of the CS repository view in openLCA, A new tool: Parameter...

openLCA.conf '25 now open to abstract submission, and sponsorship packages

openLCA Nexus: The source for LCA data sets

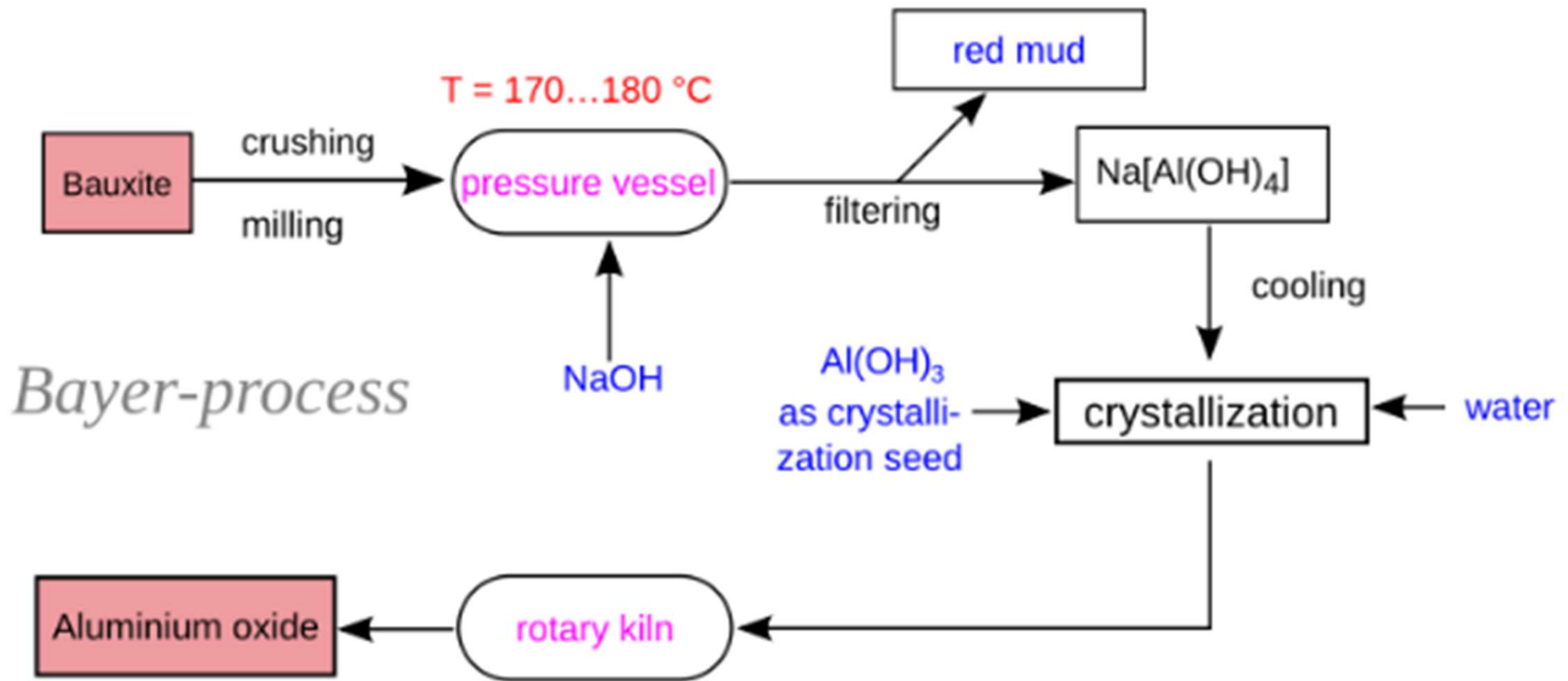
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The Bayer process flow diagram.



LCA Quantities

1 litre = approx 3 soft drink cans

Or 45 grams of aluminium

Molar mass of aluminum (Al): 26.98 g/mol

Molar mass of alumina (Al₂O₃): 101.96 g/mol

From the reaction: $2\text{Al} + 3\text{O} \rightarrow \text{Al}_2\text{O}_3$

1 mole of alumina (101.96 g) produces 2 moles of aluminium ($2 \times 26.98 \text{ g} = 53.96 \text{ g}$).

So, for 45 g of aluminium, approximately **85 grams** of alumina are required.

LCA Quantities

What are the water requirements to produce 85 g of alumina using the Bayer process?

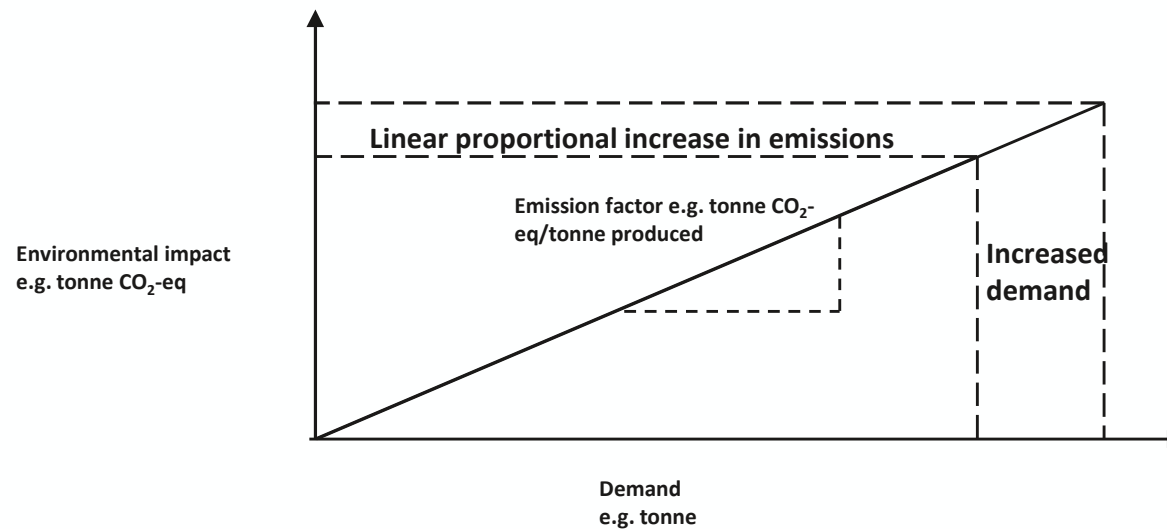
What are the diesel requirements for producing 85 grams of alumina from bauxite?

What are the bauxite requirements for producing 85 grams of alumina?

What are the electricity requirements to produce 85 g of alumina using the Bayer process?

The problem with most LCAs...

Assumes that increasing demand will result in exactly the same (historical) environmental impact

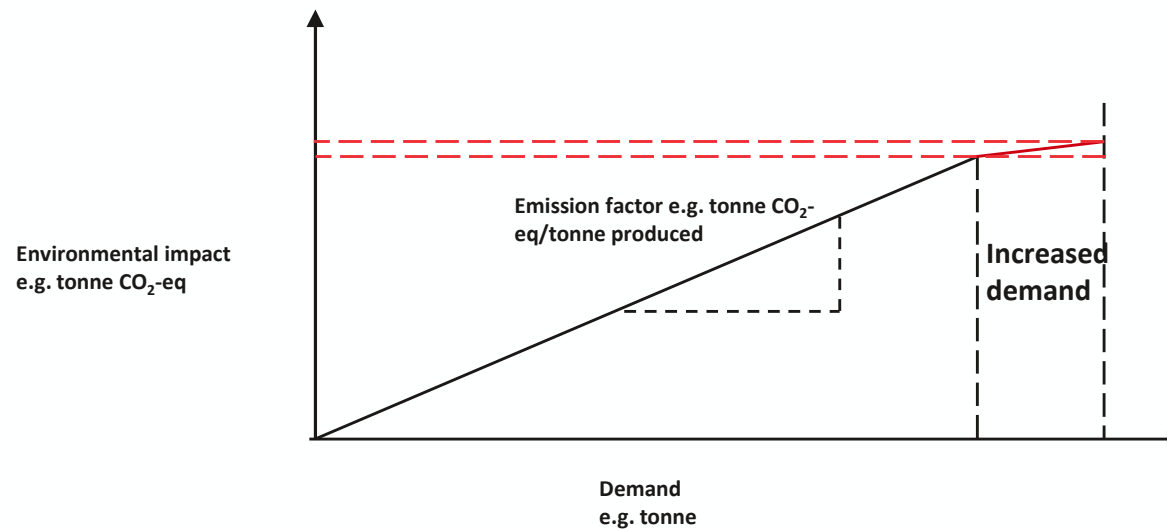


Consequential LCA

- Models the consequences from a change in demand....
- Need to also consider displacement effects in the economy
- Supply & demand
 - In a supply-constrained market, increased demand results in higher prices at the “supply margins”
 - Higher prices → someone will be ‘squeezed out’ & chooses something cheaper instead
 - The cheaper alternative can have very different environmental impact
 - Can cause ‘ripple’ of consequential effects, and unintended outcomes

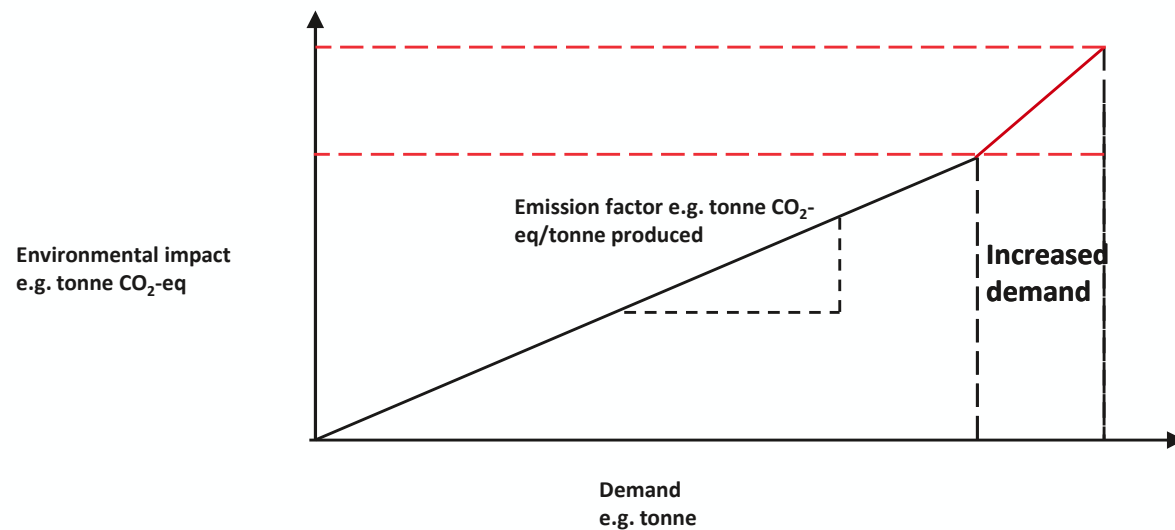
Consequential LCA

- Models the consequences from a change in demand....



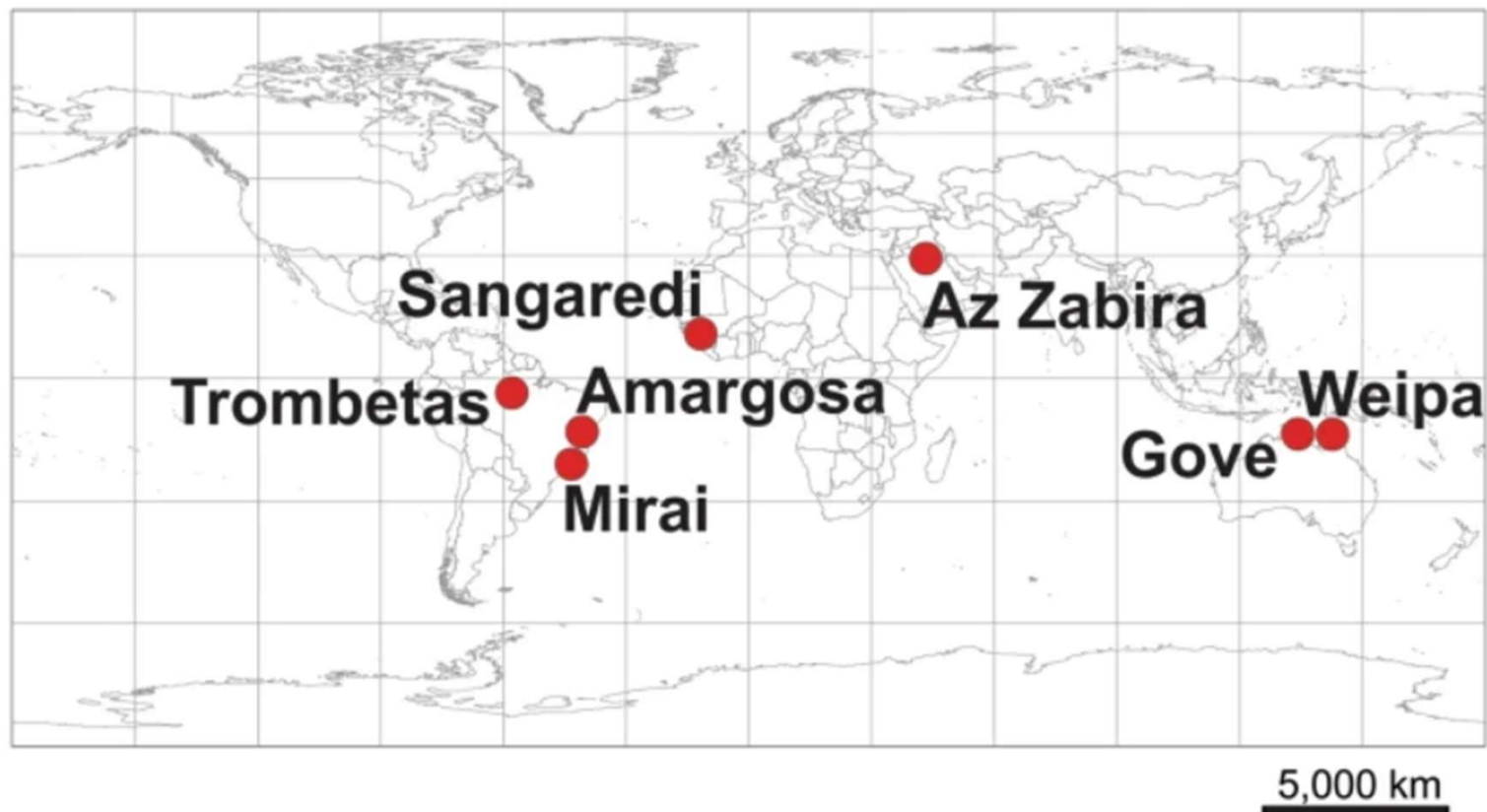
Consequential LCA

- Models the consequences from a change in demand....



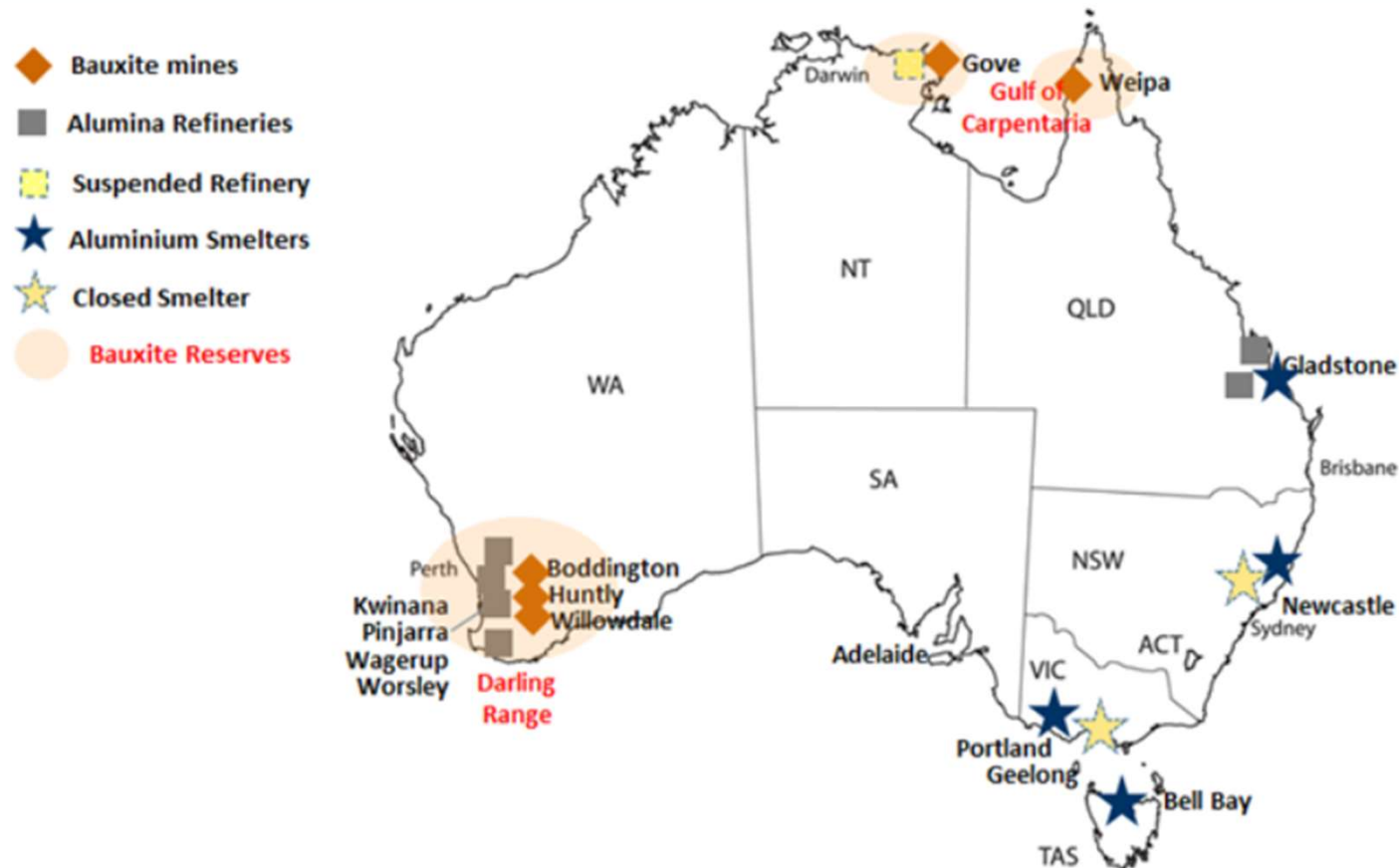
Sustainability outcomes are inter-twined with the economy

Bauxite mine sites [1]

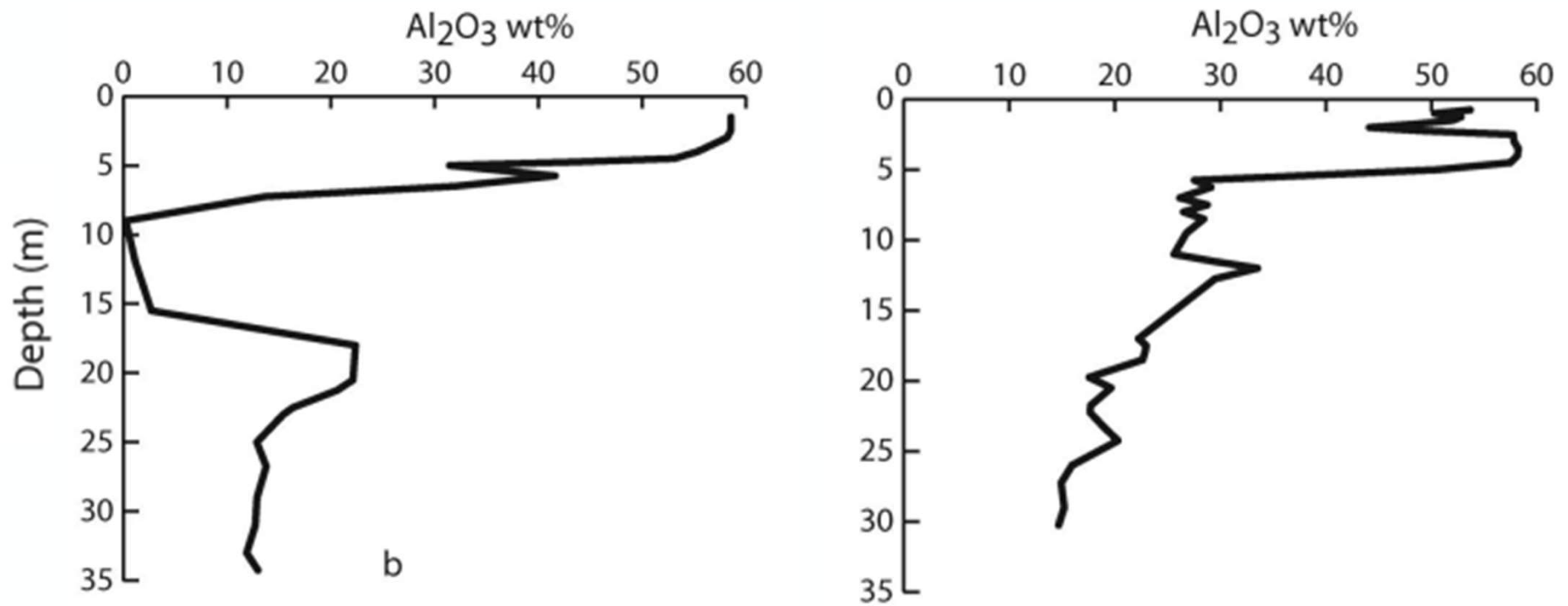


[1] Abzalov, M., & Abzalov, M. (2016). Bauxite Deposits. *Applied Mining Geology*, 411-425.

Australian Aluminium Assets



LCA Quantities – Weipa Bauxite [2]



[2] Eggleton, R. A., Taylor, G., Le Gleuher, M., Foster, L. D., Tilley, D. B., & Morgan, C. M. (2008). Regolith profile, mineralogy and geochemistry of the Weipa Bauxite, northern Australia. *Australian Journal of Earth Sciences*, 55(S1), S17-S43.

LCA Quantities – Weipa Bauxite



LCA Quantities – Weipa Bauxite [2]

What are the water requirements to produce 85 g of alumina using the Bayer process?

Answer: 2.5 to 3 tons of water per ton of alumina or **255g of water per 85g.**

What are the gas requirements for producing 85 grams of alumina from bauxite?

Answer: 250 cubic metres of gas per ton of alumina or **0.021 cubic metres of gas per 85g.**

What are the bauxite requirements for producing 85 grams of alumina?

Answer: 2.0 to 2.5 tons of bauxite per ton of alumina or **190g of bauxite per 85g.**

What are the electricity requirements to produce 85 g of alumina using the Bayer process?

Answer: 150 to 250 kWh per tonne or **0.017 kWh per 85g.**

LCA Quantities – Weipa Bauxite [2]

What are the NaOH requirements to produce 85 g of alumina using the Bayer process?

Answer: 80 to 110 kg per tonne of alumina or **8.1 g of NaOH per 85g.**

What are the CaO requirements for producing 85 grams of alumina from bauxite?

Answer: 10 to 30 kg per tonne of alumina or **1.7 g of CaO per 85g.**